

process of decryption. In the process of encryption, the information is concealed through a mathematical process using encryption key. The resultant encrypted message is scrambled so that it becomes unreadable during transmission. At the receiving end, the message can be unscrambled and decrypted into plaintext using the decryption key by the intended recipient, but not by eavesdroppers.

There are several encryption schemes like symmetric, asymmetric and hybrid. In symmetric encryption, the same key is used to encrypt and decrypt plaintext. Asymmetric encryption uses two different keys, one for encrypting and the other for decrypting. The two keys that are used are called public (encryption) and private keys (decryption).

Asymmetric encryption is usually faster. However, when using different keys, it can become difficult. Asymmetric key pairs provide better protection than a symmetric key but they are slower at encrypting/decrypting. For this reason, hybrid encryption is used, in which an asymmetric key pair is used to establish a symmetric key. So the communicating parties have the benefit of authentication without the speed penalties.

Firewall

Firewall is a system or group of systems that enforces an access control policy between two networks. It controls traffic between a private network and the Internet in order to intercept outsiders trying to break into the private network. Firewalls allow or deny protocols, ports or IP address in order to protect network and systems from attack and intentional hostile intrusion that can compromise confidentiality, cause data loss or result in denial-of-service.

Although the primary purpose of firewalls is to protect the network against unauthorised logins from outside the network, these are also sometimes deployed in the corporate world to prevent employees from going out to the Internet. In such instances, the network administrator sets a firewall so that the user can access the corporate e-mail service but not the Internet.

A firewall inspects all the inbound and outbound traffic. For this purpose, it looks at every packet that tries to enter or leave the network. If the firewall finds everything as per the access policy, it allows the packet to proceed on its way to the destination.

In most cases, the firewall records where the packet came from, where it is going and when it was seen. Firewall can also record when a connection ends.

Most firewalls come with preconfigured security policies with the provision of customising these policies for specific administrative needs. The policies can be set to block or allow data packets on the basis of their point of origin or destination or on the basis of the contents. Alerting and logging is a very important feature of a firewall. This feature allows the administrator to identify

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